

(ii) *Backward Differences*

$$y'(x_i) = \frac{y_i - y_{i-1}}{h}; \quad y'(x_i) = \frac{3y_i - 4y_{i-1} + y_{i-2}}{2h};$$

$$y''(x_i) = \frac{y_i - 2y_{i-1} + y_{i-2}}{h^2}; \quad y''(x_i) = \frac{2y_i - 5y_{i-1} + 4y_{i-2} - y_{i-3}}{h^2};$$

(iii) *Central Differences*

$$y'(x_i) = \frac{y_{i+1} - y_{i-1}}{2h}; \quad y'(x_i) = \frac{-y_{i+2} + 8y_{i+1} - 8y_{i-1} + y_{i-2}}{12h};$$

$$y''(x_i) = \frac{y_{i-1} - 2y_i + y_{i+1}}{h^2};$$

$$y''(x_i) = \frac{-y_{i+2} + 16y_{i+1} - 30y_i + 16y_{i-1} - y_{i-2}}{12h^2}$$

These formulae can be derived by using Taylor series expansion of the functions.

6.3 MAXIMUM AND MINIMUM VALUES OF A TABULATED FUNCTION

It is known that the maximum and minimum values of a function can be found by equating the first derivative to zero and solving for the variable. The same procedure can be applied to determine the maxima and minima of a tabulated function.

Consider Newton's forward difference formula

$$y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{6}\Delta^3 y_0 + \dots$$

Differentiating this with respect to p , we obtain

$$\frac{dy}{dp} = \Delta y_0 + \frac{2p-1}{2}\Delta^2 y_0 + \frac{3p^2-3p+2}{6}\Delta^3 y_0 + \dots \quad (6.25)$$

For maxima or minima $dy/dp = 0$. Hence, terminating the right-hand side, for simplicity, after the third difference and equating it to zero, we obtain the quadratic for p

$$c_0 + c_1 p + c_2 p^2 = 0, \quad (6.26)$$

where

$$\left. \begin{aligned} c_0 &= \Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 \\ c_1 &= \Delta^2 y_0 - \Delta^3 y_0 \\ c_2 &= \frac{1}{2}\Delta^3 y_0. \end{aligned} \right\} \quad (6.27)$$

and

Values of x can then be found from the relation $x = x_0 + ph$.

Example 6.7 From the following table, find x , correct to two decimal places, for which y is maximum and find this value of y .

x	y
1.2	0.9320
1.3	0.9636
1.4	0.9855
1.5	0.9975
1.6	0.9996

The table of differences is

x	y	Δ	Δ^2
1.2	0.9320		
		0.0316	
1.3	0.9636		-0.0097
		0.0219	
1.4	0.9855		-0.0099
		0.0120	
1.5	0.9975		-0.0099
		0.0021	
1.6	0.9996		

Let $x_0 = 1.2$. Then formula (6.25), terminated after second differences, gives

$$0 = 0.0316 + \frac{2p-1}{2}(-0.0097)$$

from which we obtain $p = 3.8$. Hence

$$x = x_0 + ph = 1.2 + (3.8)(0.1) = 1.58.$$

For this value of x , Newton's backward difference formula at $x_n = 1.6$ gives

$$\begin{aligned} y(1.58) &= 0.9996 - 0.2(0.0021) + \frac{-0.2(-0.2+1)}{2}(-0.0099) \\ &= 0.9996 - 0.0004 + 0.0008 \\ &= 1.0. \end{aligned}$$

6.4 NUMERICAL INTEGRATION

The general problem of numerical integration may be stated as follows. Given a set of data points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ of a function $y = f(x)$, where $f(x)$ is not known explicitly, it is required to compute the value of the definite integral

$$I = \int_a^b y \, dx. \quad (6.28)$$